Stage B: protocol verification and freeze

FINAL_V1 - Parkinson's disease detection from speech

Protocol FINAL_V1 Status: **COMPLETED** Generated: 2026-10-20T09:37:38Z

Protocol status

- No frozen protocol file was supplied. A protocol was drafted by the agent and frozen after the owner's instruction (D-000): FINAL_V1_PROTOCOL.md and configs/FINAL_V1/protocol.json.
- Freeze time 2026-10-20T09:37:38Z UTC. Protocol JSON SHA-256 5535d1a440a57005....
- Split manifest SHA-256 31c40ed902165aff....

Design

- Speaker-disjoint stratified 5-fold outer CV (seed 42) and speaker-disjoint 3-fold inner CV for tuning and feature selection.
- 12 representations (acoustic, WavLM-large, wav2vec2-large-lv60, XLS-R 300M, 8 hybrids) x 4 models (LR, linear SVM, RBF SVM, XGBoost); acoustic features also with forward and backward selection (k=10). 56 conditions, 280 outer-fold runs.
- Primary metrics: pooled speaker-level AUPRC and AUROC; fixed threshold 0.5; inner objective speaker-level AUPRC; bootstrap CIs over speakers.

Compliance check and open items

- All automatic compliance checks passed.
- Unresolved methodological decision: FWL is BLOCKED (PROTOCOL_DECISION_REQUIRED) because no purpose, response, covariates, statistic or stopping rule exists. It is recorded, not improvised.
- No implementation detail is open. Historical scores were not used for any design choice.

Table 1: Conditions by representation

Representation	Components	Dim	Conditions
AC	AC	40	12
WLM	WLM	1024	4
W2V	W2V	1024	4
XLSR	XLSR	1024	4
H1	WLM+AC	1064	4
H2	W2V+AC	1064	4
H3	XLSR+AC	1064	4
H4	WLM+W2V	2048	4
H5	WLM+XLSR	2048	4
H6	W2V+XLSR	2048	4
H7	WLM+W2V+XLSR	3072	4
H8	WLM+W2V+XLSR+AC	3112	4